# Signal Detection and Metacognition

Sensitivity, bias and confidence, applied to LLM introspection

Measuring AI Self-Reports

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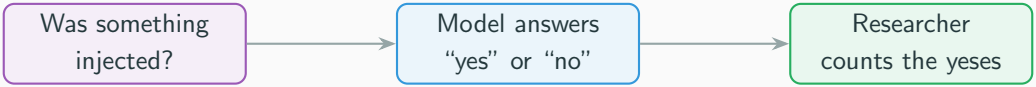
# Part 1

Sensitivity and saying yes

# The running example

Lindsey (2025) runs a *concept injection* experiment:

1. Take the model's internal activation pattern for a word, such as *bread*.
2. Add it to the model's activations on some trials, and leave the activations alone on others.
3. Ask: *Do you detect an injected thought?*




Signal detection theory is the standard way to read a pile of yes/no answers. It splits a rate of "yes" into two things: how well the model tells injection from no injection, and how willing it is to say yes.

# Four outcomes

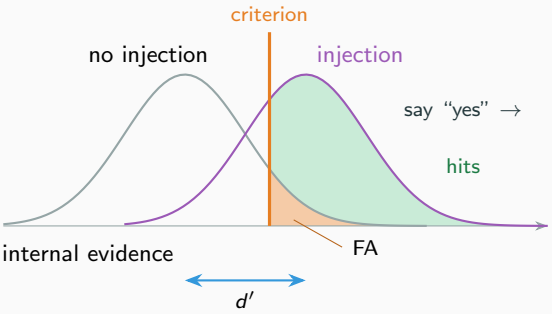
Every trial lands in one of four cells:

| | **Model says "yes"** | **Model says "no"** |
|---|---|---|
| **Injected** | **Hit** *noticed a real signal* | **Miss** *missed a real signal* |
| **Not injected** | **False alarm** *"yes" to nothing* | **Correct rejection** *"no" to nothing* |

The **hit rate** $H$ is the share of "yes" on injected trials. The **false-alarm rate** $F$ is the share of "yes" on trials with nothing injected.

Each row needs its own trials, and both rows are needed.

# Two bell curves and a line




On each trial the model has some amount of internal evidence. It varies from trial to trial, so each condition gives a bell curve.

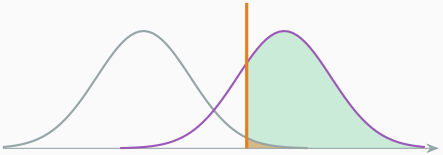
**Sensitivity** ($d'$): how far apart the two curves are.

**Criterion**: the line. Evidence to the right of it gets a "yes".

Green area: hits. Orange area: false alarms (FA).

# Same hit rate, different observers

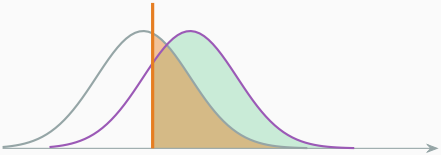
**High sensitivity, strict criterion**



Curves far apart; line far right.

Almost no false alarms.

**Low sensitivity, lax criterion**



Curves overlap; line far left.

Many false alarms (orange).

Both observers say "yes" on the same share of injected trials. One is sharp and cautious; the other barely tells the conditions apart and says yes readily.

A hit rate alone cannot tell them apart. Only the false-alarm rate can.

# Sensitivity and criterion as numbers

Under the equal-variance Gaussian model:

$$d' = z(H) - z(F) \qquad\qquad c = -\tfrac{1}{2}\left[z(H) + z(F)\right]$$

$z$ turns a rate into a position on the bell curve, measured in standard deviations (the inverse of the standard normal cumulative distribution function).

$d'$: distance between the curves
*stays the same if only the line moves*

$c$: position of the line
*the tendency to say yes*

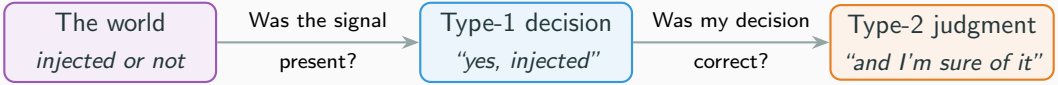
Percentage correct mixes the two. $d'$ stays constant when only the criterion changes (Fleming & Lau 2014).

# Part 2

Confidence about one's own answers

# Type-1 and type-2 tasks

Metacognition research applies the same split one level up, to confidence.




A **type-1** task asks about the world. A **type-2** task asks about the observer's own type-1 decision.

The type-2 score is checked against whether the observer was right, not against the state of the world (Fleming & Lau 2014).

# The confidence-accuracy table

Same four cells as before, one level up. Rows are now *was I right*, columns are *how sure was I*:

| | **High confidence** | **Low confidence** |
|---|---|---|
| **Correct** | **Type-2 hit**<br>*right and knew it* | **Type-2 miss**<br>*right but unsure* |
| **Incorrect** | **Type-2 false alarm**<br>*wrong but sure* | **Type-2 correct rejection**<br>*wrong and unsure* |

Good metacognition fills the green cells: confidence is high when the decision was right and low when it was wrong.

# Sensitivity, bias and efficiency

Three quantities are defined on this table (Fleming & Lau 2014):

**Sensitivity**
Does confidence separate my right answers from my wrong ones?

**Bias**
How confident am I overall, right or wrong?

**Efficiency**
Sensitivity relative to how well I do the task itself

Bias is also called over- or underconfidence, or calibration. Sensitivity is also called metacognitive accuracy or discrimination.

A subject can be confident overall and still have poor sensitivity, if confidence is equally high on correct and incorrect trials.

# Correlations and type-2 $d'$

Fleming & Lau 2014 review the obvious measures of metacognitive sensitivity. Two families fail:

**Phi and gamma**

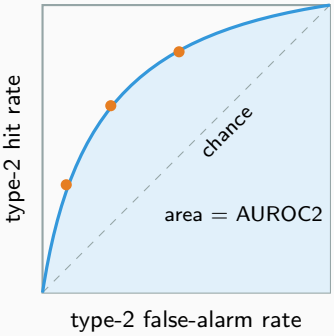
Correlate confidence with accuracy. Both move with bias: a modest subject who rates every trial low gets a low correlation, for reasons that have nothing to do with sensitivity.

**Type-2 $d'$**

Runs the $d'$ formula on the type-2 table. That assumes equal-variance bell curves for correct and incorrect trials. Galvin et al. (2003) showed they are unequal and non-Gaussian, so the measure is confounded with bias.

Both fail the same way: a change in how confident the subject is overall shows up as a change in sensitivity.

# AUROC2




Treat each confidence level as a criterion. Each one gives a point: a type-2 hit rate and a type-2 false-alarm rate (orange dots).

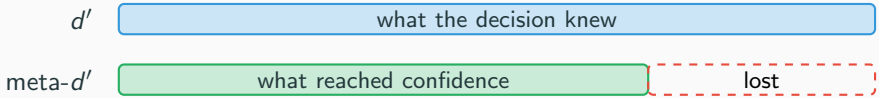
Join the points into a curve. The area under it is **AUROC2**. The diagonal is confidence that says nothing about correctness.

**Good:** free of bias.

**Problem:** it rises with task performance. The same subject scores higher on an easy task than on a hard one.

# Meta-$d'$ and the M-ratio

**Meta-$d'$** (Maniscalco & Lau 2012): the $d'$ that an ideal observer, one who uses all the evidence, would need to produce the confidence performance actually seen. Same units as $d'$, so the two compare directly.




**M-ratio** $=$ meta-$d'/d'$: how much of what you know reaches your confidence.

**M-ratio** $= 1$: a metacognitively ideal observer; all the evidence reached confidence.

**M-ratio** $= 0.7$: 30% of the evidence was lost on the way to confidence (Fleming & Lau 2014).

Michel (2023) calls the M-ratio the gold standard for metacognitive efficiency. When $d'$ is small the ratio is unstable, and meta-$d' - d'$ or the log ratio is used instead.

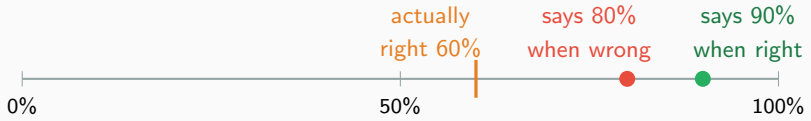
# Part 3

Probabilities and awareness

# The Brier decomposition

A stated probability gives bias an absolute scale: a stated 70% can be checked against the proportion correct. The Brier score, the mean squared gap between stated probability and outcome, splits as

$$PS = \underbrace{O}_{\text{outcome variance}} + \underbrace{C}_{\text{calibration}} - \underbrace{R}_{\text{resolution}}$$




**Calibration** (bias): poor; both confidence levels sit far above 60%. **Resolution** (sensitivity): good. Right answers always get more confidence than wrong ones (Fleming & Lau 2014).

Because $O$ depends on accuracy, the Brier score moves with task performance.

# Three senses of calibration

Papers on LLM introspection use "calibration" in more than one sense:

| Sense | Meaning | Where it appears |
|---|---|---|
| **Metrological** | Relating an instrument's indications to values of the measurand, against a standard | Metrology |
| **Probability** | Stated confidence agrees with proportion correct; the $C$ term of the Brier score | Kadavath et al.'s expected calibration error; Ferrara's Brier score |
| **Discriminatory** | Confidence rank-orders outcomes; metacognitive sensitivity under another name | Moran and Whiting, Def. 3.1 |

Kadavath et al. (2022) state the difference between the second and third: the area under the curve measures discrimination and ignores calibration.

# Michel's four conditions

Michel (2023) separates **validity** (does the procedure measure the intended attribute?) from **accuracy** (how well?), and states four conditions:

| Condition | Content | Guarantees |
|---|---|---|
| Consciousness-selectivity | Unconscious sensory activity is a source of metacognitive inefficiency | **Validity** |
| Optimal metacognition | Metacognition is optimal for decisions based on conscious activity | **Accuracy** |
| No overflow | All conscious activity is available for metacognitive decisions | **Accuracy** |
| No hallucination | Activity driving false alarms is not conscious | **Accuracy** |

Other sources of inefficiency are confounds. Controls: remove them (*ceteris absentibus*), equate them across conditions (*ceteris paribus*), or make them negligible (*ceteris neglectis*).

# Part 4

Language models

# Concept injection as a type-1 task

The stimulus is internal, but the question is type-1: was a signal present?

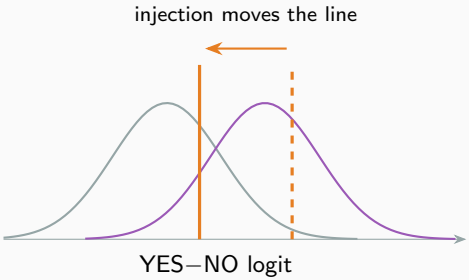
| Hit rate | False-alarm rate |
| --- | --- |
| *"yes" on injection trials* | *"yes" with nothing injected* |

Lindsey (2025) reports **no false positives in 100 control trials**. That is good news for the model and a problem for the arithmetic:

$$F = 0 \;\Rightarrow\; z(F) = -\infty \;\Rightarrow\; d' = z(H) - z(F) \text{ is infinite.}$$

So $d'$ depends on the **correction chosen for the empty cell**. Ferrara (2026) uses a $1/(2N)$ correction against a paired sham trial and reports a pooled $d' = 0.0039$.

# Injection moves the criterion



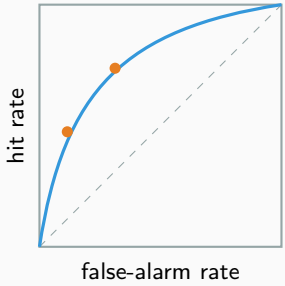

Hahami et al. (2025) read the YES-minus-NO logit in Llama-3.1-8B-Instruct. They apply the same injection to control questions whose answer is no.

The rise on the detection question and the rise on the controls correlate at $\mathbf{r = 0.999}$.

In signal detection terms, the injection shifts the **criterion**: the model becomes more likely to say yes to any question.

Lindsey's check on unrelated questions in Claude models finds no increase in yes answers.

# A logit as a decision variable




**Rating scale:** a few criteria, so a few points on the ROC curve (orange dots).

**Logit difference:** continuous. Every possible threshold is a criterion, so the full receiver operating characteristic curve (blue) comes from one run, without asking for ratings.

Hahami et al. read every answer this way.

# Confidence papers as type-2 tasks

Three papers score a model's confidence against its own correctness:

**Kadavath et al. 2022**
Probability calibration and AUROC

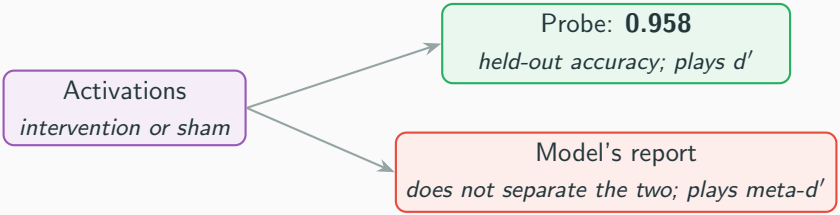
**Ackerman 2025**
Partial correlation between delegation and correctness: the phi family

**Moran & Whiting 2026**
Thresholded latent normal variable, one threshold per model

Ackerman's measure is in the phi family, which moves with bias. Moran and Whiting's threshold separates a model's yes rate from the shared signal; they state that the normality assumption is presumed rather than tested.

# A probe as the ideal observer

Meta-$d'$ compares confidence with the evidence the decision had. The closest LLM version compares the model's report with a probe trained on the same activations.




Ferrara (2026), Qwen2.5-7B-Instruct. Ferrara defines the ratio of report margin to probe margin as an M-ratio analog.

In plain terms: the information is in the activations, and it does not reach the report.

# Part 5

Limits

# Where the analogy stops

- **Items, not internal noise.** Signal detection's standard design holds the stimulus constant, so variation in answers reflects noise inside the observer (Fleming & Lau 2014). Many LLM experiments sample at temperature 0, where an item always gets the same answer. Variation then comes from items, and $d'$ describes distributions over items.
- **One subject.** Human studies estimate efficiency per subject and compare subjects. A model sampled many times is one subject, and differences between models mix architecture, training and prompt format.
- **Type-1 and type-2 can merge.** Meta-$d'$ rests on a model of the type-1 decision. In an LLM the confidence prompt can make the model solve the item again. Moran and Whiting report that reasoning-trained models often do, which turns a type-2 judgment into a second type-1 attempt.

# Summary

| Idea | Question for an introspection experiment | Card stage |
|---|---|---|
| Hits and false alarms | Is the yes rate without injection reported? | Instrument |
| $d'$ versus criterion | Is a yes-bias separated from sensitivity? | Measurement model |
| Empty cells | Which correction is used when $F = 0$? | Uncertainty |
| Type-1 versus type-2 | Is the task about the world or the model's answer? | Measurement model |
| Sensitivity, bias, efficiency | Which of the three does the measure track? | Measurement model |
| AUROC2 and Brier score | Does the number move with task performance? | Uncertainty |
| Three calibrations | Which sense of calibration is meant? | Calibration |
| Meta-$d'$ and the probe | Is the report compared with a probe? | Comparability |
| Constant stimuli | Does variation come from items or the model? | Validity conditions |

# Reading

- Fleming & Lau, How to measure metacognition (2014): sensitivity, bias and efficiency; bias-free measures; the Brier decomposition.
- Maniscalco & Lau, A signal detection theoretic approach for estimating metacognitive sensitivity from confidence ratings (2012): meta-$d'$.
- Maniscalco, type-2 SDT code: fitting meta-$d'$.
- Michel, Confidence in consciousness research (2023): validity versus accuracy; conditions and confounds.
- Michel, The old and new criterion problems (2023): response bias in type-1 reports.
- Signal detection and metacognition: the site page this deck follows.